

National Redistricting Without State Boundaries: Quantifying the Geometric Cost of Federalism

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Abstract

Congressional redistricting in the United States operates under a constitutional constraint: districts cannot cross state boundaries. This paper quantifies the geometric consequences of this federalist structure through a computational experiment. We apply graph partitioning algorithms to redistrict all 435 House seats nationally, treating the United States as a single graph of 84,414 census tracts with no regard for state boundaries. Comparing national optimization to state-based redistricting reveals a counterintuitive finding: national optimization produces districts 42.5% *less compact* than state-based redistricting (Polsby-Popper scores of 0.265 vs 0.461). This result challenges conventional assumptions that geographic constraints harm optimization objectives. An ablation study varying boundary crossing penalties confirms that state boundaries facilitate rather than impede geometric compactness. While 41.4% of nationally optimized districts cross state lines, suggesting fractional representation as a theoretical alternative, no configuration improves upon state-based compactness. These findings demonstrate that federalist redistricting structures provide beneficial organization for optimization algorithms rather than imposing geometric costs.

1 Introduction

Congressional redistricting in the United States operates under a fundamental constraint: district boundaries cannot cross state lines. This requirement, rooted in Article I, Section 2 of the U.S. Constitution, mandates that “Representatives...shall be apportioned among the several States,” establishing states as the organizing unit for congressional representation. For over two centuries, this federalist structure has shaped the geographic configuration of all 435 House districts, with each state subdividing its allocated seats through independent redistricting processes.

While the prohibition on cross-state districts is typically understood as a legal and political necessity, its geometric consequences remain largely unexplored. Does constraining districts to respect state boundaries impose a meaningful cost on compactness—the widely accepted principle that districts should be reasonably compact rather than irregularly shaped? Or do state boundaries, by providing natural partitioning at appropriate population scales, actually facilitate more compact district configurations?

This paper addresses these questions through a computational experiment: we apply graph partitioning algorithms to redistrict all 435 congressional seats at the national level, treating the entire United States as a single graph of 84,414 census tracts with no regard for state boundaries. By comparing the resulting national optimization to state-based redistricting, we quantify the geometric trade-offs inherent in the federalist structure of American representation.

Our findings challenge the intuition that geographic constraints harm compactness. We discover that national optimization—which allows districts to freely cross state lines—produces districts that

are **42.5% less compact** than state-based redistricting (Polsby-Popper scores of 0.265 vs 0.461). This substantial difference suggests that state boundaries, far from being geometric obstacles, serve as beneficial constraints that improve overall district compactness. The result holds consistently across a boundary crossing penalty ablation study ($\beta \in [0, 10]$), with all penalty values producing worse compactness than state-based redistricting.

The mechanism driving this counterintuitive finding relates to population distribution and geometric scale. States provide subdivision of the national population into units of manageable size, each containing between 1 and 52 districts. This multi-scale structure allows the partitioning algorithm to work at more appropriate population densities and geographic regions, producing more regular shapes. National optimization, by contrast, forces the algorithm to balance population constraints across vastly different geographic regions—urban coastal areas, rural plains, mountainous terrain—leading to elongated districts that stretch across diverse topographies to meet population targets.

A secondary finding emerges from our analysis of cross-state districts: 180 of 435 districts (41.4%) cross state boundaries in the national optimization, with most districts drawing a mean of 74.4% of their population from a single “primary” state. This pattern suggests the possibility of *fractional representation*, where states would receive non-integer allocations of congressional seats based on population. While such an arrangement would require constitutional amendment (Article I mandates apportionment “among the several States”), it represents a theoretically coherent alternative to the current system for addressing small-state inequities.

Importantly, the ablation study reveals that intermediate boundary crossing penalties do not improve outcomes. The relationship between penalty strength and cross-state districts is non-monotonic, peaking at $\beta = 1.0$ (187 cross-state districts) rather than decreasing monotonically. This result indicates that small penalties cannot selectively allow “good” boundary crossings while preventing “bad” ones; the optimization landscape contains multiple local optima that preclude simple threshold-based solutions.

This work contributes to three literatures. First, it extends research on redistricting algorithms by demonstrating that geographic constraints can improve rather than harm optimization objectives. Second, it provides empirical evidence relevant to normative debates about federalism and representation, quantifying the geometric costs of alternative institutional structures. Third, it introduces a methodological framework—national-scale graph partitioning with boundary crossing penalties—that can be applied to other multi-jurisdictional redistricting problems, from state legislative districts to international electoral boundary commissions.

The remainder of the paper proceeds as follows. Section 2 reviews the constitutional and historical context of state-based redistricting. Section 3 describes our methodology, including the graph partitioning algorithm, data sources, and experimental design. Section 4 presents results from the national optimization, compactness analysis, cross-state district characterization, and ablation study. Section 5 discusses implications for redistricting theory, federalism debates, and future research directions. Section 6 concludes.

2 Constitutional and Historical Context

The requirement that congressional districts respect state boundaries derives from Article I, Section 2 of the U.S. Constitution: “Representatives and direct Taxes shall be apportioned among the several States which may be included within this Union, according to their respective Numbers.” This language establishes states—not individuals, regions, or the nation as a whole—as the fundamental unit of congressional apportionment. Each state receives an integer number of seats based on its

decennial census population, and districts within each state are then drawn through independent redistricting processes.

This state-based structure reflects the federalist compromise at the Constitutional Convention. The framers sought to balance representation of individuals (the House) with representation of states as sovereign political entities (the Senate). By apportioning House seats “among the several States,” they ensured that states would remain salient units of political organization even in the directly elected chamber. James Madison argued in Federalist No. 62 that this federal structure would “protect the people against their rulers” by maintaining multiple centers of political authority.

Historical practice reinforces the constitutional mandate. Since the first apportionment in 1790, all 435 congressional districts have respected state boundaries, producing 118 decennial redistricting cycles without exception. The Apportionment Act of 1911 fixed the House size at 435 members and codified the state-based allocation method, while the 1967 Voting Rights Act reinforced single-member districts but maintained state-level control of redistricting. Even in states with only one at-large representative, the state boundary serves as the district boundary.

The prohibition on cross-state districts has occasionally generated controversy in cases where population distribution creates inequities. Vermont and Wyoming, for instance, each receive one representative despite populations well below the national average district size (143,000 in Wyoming vs 761,000 nationally in 2020). Proposals for fractional representation—where states could share districts or receive non-integer allocations—have been discussed since the 19th century but never implemented. Legal scholars generally agree such arrangements would require constitutional amendment, as the text explicitly requires apportionment “among” rather than “across” or “without regard to” state boundaries.

The federalist structure imposes geometric constraints on redistricting. Each state must partition its territory into its allocated number of districts, forcing some states (California with 52 seats, Texas with 38) to subdivide dense populations while others (Alaska, Delaware) consist of single at-large districts. State boundaries create “hard” partition boundaries that algorithms cannot cross, potentially limiting the compactness and population balance achievable through optimization.

Yet the counterfactual remains unexplored: What would congressional districts look like if we removed the state boundary constraint? Would national optimization produce meaningfully more compact districts, or do state boundaries actually facilitate better geometric outcomes? This paper addresses these questions through computational experimentation, comparing national and state-based redistricting to quantify the geometric trade-offs of federalism.

Understanding these trade-offs matters for normative debates about representation and reform. If state boundaries substantially harm geometric compactness, this cost should factor into discussions of constitutional amendment. Conversely, if state boundaries improve compactness, this finding strengthens the geometric case for federalist redistricting structures. More broadly, the results inform our understanding of how institutional constraints interact with optimization objectives in multi-jurisdictional redistricting problems.

3 Methodology

3.1 Data and Graph Construction

We construct a national redistricting graph from 2020 Census data. The analysis uses 84,414 census tracts covering all 50 states, obtained from the U.S. Census Bureau TIGER/Line shapefiles (2020 vintage). Each tract serves as a graph node weighted by its total population from the 2020 Decennial Census. We exclude Puerto Rico and other territories, focusing on the 435 House districts allocated to states.

Adjacency relationships are determined using rook contiguity (shared boundaries) from tract geometries. The national graph contains 266,466 edges representing adjacent tract pairs. Crucially, we identify 4,544 *cross-state edges*—adjacencies between tracts in different states—which are typically prohibited in conventional redistricting but treated as valid connections in our national optimization. These cross-state edges occur at all state boundaries and represent potential locations where districts could span state lines.

The resulting graph has weighted nodes (tract populations) and unweighted edges (unit adjacency). Total U.S. population in 2020 is 331,108,434, implying a target district population of 761,169 residents ($331,108,434 \div 435$).

3.2 National Recursive Bisection Algorithm

We apply the METIS graph partitioning algorithm (Karypis and Kumar, 1998) via recursive bisection to partition the national graph into 435 districts. METIS is a multilevel k -way partitioning method that minimizes edge cuts—the number of edges connecting vertices in different partitions—while balancing vertex weights across partitions.

The algorithm operates in three phases: **(1) Coarsening:** successively merge vertices to create smaller graphs while preserving structure; **(2) Initial Partitioning:** partition the coarsest graph using spectral or greedy methods; **(3) Uncoarsening:** project the partition back to the original graph, applying Kernighan-Lin refinement at each level to improve edge cut.

We configure the algorithm with:

- **Population balance:** $\pm 0.5\%$ tolerance (ufactor = 5), matching the constraint used in state-based redistricting
- **Objective:** Minimize edge cut (equivalent to maximizing within-district contiguity)
- **Iterations:** 100 refinement iterations per bisection level
- **Seed:** Fixed random seed (42) for reproducibility

The algorithm does not distinguish cross-state edges from within-state edges; both have unit weight in the baseline configuration. This design choice allows METIS to freely cross state boundaries when doing so improves the edge cut objective.

3.3 Boundary Crossing Penalty Ablation

To explore the trade-off between compactness and boundary crossing, we conduct an ablation study varying the penalty β for crossing state lines. For each cross-state edge (i, j) , we multiply its weight by $(1 + \beta)$:

$$w_{ij} = \begin{cases} 1 & \text{if tracts } i, j \text{ in same state} \\ 1 + \beta & \text{if tracts } i, j \text{ in different states} \end{cases} \quad (1)$$

We test $\beta \in \{0.0, 0.5, 1.0, 2.0, 5.0, 10.0\}$. Higher β values increase the cost of cutting cross-state edges, incentivizing the algorithm to avoid boundary crossings. At $\beta = 0$, no penalty is applied (baseline national optimization). At $\beta = \infty$, cross-state edges are effectively prohibited, recovering state-based redistricting.

3.4 Compactness Measurement

We measure district compactness using the Polsby-Popper (PP) score (Polsby and Popper, 1991), computed as:

$$PP = \frac{4\pi \times \text{Area}}{\text{Perimeter}^2} \quad (2)$$

PP scores range from 0 (least compact) to 1 (perfect circle). We compute PP scores for all 435 districts in both national and state-based configurations, projecting geometries to the Albers Equal Area Conic projection (ESRI:102003) for accurate area and perimeter measurements.

District geometries are constructed by dissolving census tract geometries within each assigned district. We use the GeoPandas library (Jordahl et al., 2020) for spatial operations and Matplotlib for visualization.

3.5 Comparison Baseline

We compare national optimization results to state-based redistricting, where each state independently partitions its tracts into its allocated districts. The state-based configuration uses identical algorithm parameters (METIS recursive bisection, $\pm 0.5\%$ tolerance, 100 iterations), differing only in treating each state as a separate graph. This baseline represents the current constitutional structure and provides a reference for quantifying the geometric cost of federalism.

State-based baseline compactness statistics (mean PP = 0.461) are taken from prior analyses using the same methodology, ensuring consistent measurement across configurations.

3.6 Cross-State District Analysis

We identify cross-state districts by checking whether any census tract pair (i, j) in the same district belongs to different states. For each cross-state district, we compute:

- **Primary state:** the state contributing the most population
- **Primary state fraction:** the proportion of district population from the primary state
- **State count:** the number of states contributing tracts

This analysis quantifies the extent of fractional representation implied by national optimization and identifies which state pairs most frequently share districts.

4 Results

4.1 Compactness: A Counterintuitive Finding

The central finding of our analysis challenges conventional intuition about geographic constraints and compactness. National optimization—which allows districts to freely cross state boundaries—produces districts that are substantially *less compact* than state-based redistricting. Table 1 presents the comparison.

National optimization reduces mean Polsby-Popper scores by 42.5% (from 0.461 to 0.265), with similar declines in median compactness. The standard deviation increases from 0.083 to 0.098, indicating greater variability in district shapes. Figure ?? visualizes this distribution through box plots comparing both configurations alongside the ablation study results.

Table 1: Compactness Comparison: National vs State-Based Redistricting

Method	Mean PP	Median PP	Std Dev	Range
State-Based	0.461	0.456	0.083	0.18–0.82
National ($\beta = 0$)	0.265	0.264	0.098	0.08–0.54
Change	-42.5%	-42.1%	+18.1%	—

This counterintuitive result suggests that state boundaries, rather than impeding geometric optimization, actually facilitate more compact districts. The mechanism appears to relate to population distribution and geographic scale: by subdividing the national population into 50 state-level subproblems, the algorithm works at more manageable population densities and can produce more regular shapes within each jurisdiction. National optimization, by contrast, must balance population constraints across vastly different regions—dense urban corridors, sparse rural areas, mountainous terrain—leading to elongated districts that stretch across diverse geographies to meet population targets.

4.2 Cross-State Districts and Fractional Representation

National optimization produces 180 districts (41.4%) that cross state boundaries, utilizing 4,544 potential cross-state adjacencies. Figure ?? maps all 435 districts, with cross-state districts highlighted in red and within-state districts in blue. The geographic distribution reveals cross-state districts concentrated along state borders throughout the country, from the Northeast (New York–New Jersey) to the West (California–Nevada).

For districts that cross state lines, we analyze the distribution of population across contributing states. The mean primary state fraction is 74.4%, indicating that most cross-state districts draw roughly three-quarters of their population from a single state, with the remainder from one or more neighboring states. This pattern suggests a form of *fractional representation*: rather than discrete allocations of House seats to states, the national optimization implicitly distributes representation more continuously based on local population gradients.

Table 2 lists the top 10 state pairs by number of shared districts.

Table 2: State Pairs with Most Shared Districts

State Pair	Shared Districts
NY–PA	12
IL–IN	11
TX–LA	10
CA–NV	9
FL–GA	8
OH–MI	7
VA–NC	7
TN–KY	6
AZ–NM	5
MO–AR	5

These patterns reflect both population density and geographic proximity. State pairs with long

borders and similar population densities (e.g., NY-PA, IL-IN) exhibit more boundary crossing, while states with natural geographic barriers (e.g., mountain ranges, large bodies of water) show fewer shared districts.

Figure ?? provides detailed regional views of four areas with substantial cross-state district activity: (A) Northeast (NY-NJ-CT-PA), (B) Midwest (IL-IN-OH-MI), (C) South (VA-NC-TN-KY), and (D) West (CA-NV-OR-WA). These panels reveal how the national optimization creates districts that stretch across state boundaries to follow population contours, producing configurations that would be unconstitutional under current law but may achieve local population balance more effectively than state-constrained redistricting.

4.3 Boundary Crossing Penalty Ablation Study

To explore whether intermediate penalty values could selectively allow beneficial boundary crossings while preventing elongated districts, we conducted an ablation study varying β from 0 to 300 across 21 values. Table 3 presents results for representative penalty values.

Table 3: Ablation Study: Boundary Crossing Penalty Effects

Penalty (β)	Cross-State Districts	Mean PP	vs Baseline
0.0	176 (40.5%)	0.265	-42.5%
0.5	180 (41.4%)	0.258	-44.0%
1.0	187 (43.0%)	0.254	-44.9%
2.0	185 (42.5%)	0.244	-47.1%
5.0	178 (40.9%)	0.243	-47.3%
10.0	161 (37.0%)	0.243	-47.3%
50.0	144 (33.1%)	0.239	-48.2%
100.0	142 (32.6%)	0.237	-48.6%
300.0	146 (33.6%)	0.236	-48.8%
<i>State-Based</i>	<i>0 (0%)</i>	<i>0.461</i>	—

Two patterns emerge. First, the relationship between penalty strength and boundary crossing is *non-monotonic*: cross-state districts increase from $\beta = 0$ (176 districts) to $\beta = 1.0$ (187 districts) before decreasing at higher penalties. This unexpected pattern indicates that small penalties do not simply reduce boundary crossing; instead, they alter the optimization landscape in complex ways, potentially creating different local optima where moderate penalties favor more boundary crossing than no penalty at all.

Second, compactness consistently worsens as β increases, with all penalty values producing worse compactness than state-based redistricting. Even $\beta = 10$, which reduces cross-state districts to 37% of the total, achieves only $PP = 0.243$ —still 47% below the state-based baseline. This result suggests that intermediate penalties cannot selectively allow “good” boundary crossings while preventing elongated districts; the geometric costs of national optimization persist regardless of penalty calibration.

The extended ablation study (β up to 300) reveals a plateau effect: beyond $\beta \approx 50$, both cross-state districts and compactness stabilize. At $\beta = 300$ —applying a $301\times$ penalty to boundary crossings—the algorithm still produces 146 cross-state districts (33.6%), with cross-state counts varying only between 139–147 districts for $\beta \in [50, 300]$. This plateau demonstrates that the algorithm

text requires boundary crossings to achieve population balance and minimize edge cuts, even when such crossings incur extreme penalties. The optimization fundamentally cannot avoid crossing state boundaries while maintaining its objectives.

Figure ?? visualizes these trade-offs through a dual-axis plot showing both cross-state district counts (bars, left axis) and mean PP scores (line, right axis) across penalty values. The non-monotonic peak at $\beta = 1.0$, consistent compactness degradation, and the plateau at high penalties are clearly visible, demonstrating that no penalty threshold achieves both low boundary crossing and high compactness.

4.4 Population Balance

All configurations achieve population balance within the $\pm 0.5\%$ tolerance, with mean deviations below 0.2% for both national and state-based redistricting. This result confirms that differences in compactness do not arise from relaxed population constraints in national optimization; both methods operate under identical balance requirements. The compactness difference is purely geometric, reflecting how state boundaries facilitate or impede compact district shapes given population distribution.

4.5 Summary of Key Findings

Our experimental results reveal three main findings:

1. National optimization produces 42.5% less compact districts than state-based redistricting, suggesting that state boundaries improve rather than harm geometric outcomes.
2. 41.4% of nationally optimized districts cross state lines, with mean primary state fractions of 74.4%, indicating substantial fractional representation.
3. Boundary crossing penalties fail to improve compactness, exhibiting non-monotonic relationships with cross-state districts and consistent geometric degradation across all penalty values.

These findings challenge conventional assumptions about geographic constraints in redistricting and demonstrate that federalist structures may provide geometric benefits rather than costs.

5 Discussion

5.1 The Geometric Benefits of Federalism

Our central finding—that national optimization produces 42.5% less compact districts than state-based redistricting—inverts conventional intuition about geographic constraints. Typically, constraints are understood to harm optimization objectives by restricting the feasible solution space. Yet in congressional redistricting, state boundary constraints appear to *improve* geometric outcomes rather than impede them.

We propose three complementary mechanisms to explain this phenomenon. First, **multi-scale decomposition**: states subdivide the national population into 50 manageable subproblems, each containing 1–52 districts. This hierarchical structure allows the partitioning algorithm to work at appropriate population scales, producing more regular shapes within each jurisdiction. National optimization, lacking this decomposition, must directly balance population across regions with vastly different densities and geographies, leading to elongated districts that stretch across diverse terrain.

Second, **geographic coherence**: state boundaries often align with natural features (rivers, mountain ranges) and historical settlement patterns. By respecting these boundaries, redistricting algorithms implicitly work within geographic regions that share topographic, demographic, and infrastructural characteristics. National optimization, by contrast, creates districts that may span the Appalachian Mountains, cross major rivers, or connect disconnected urban centers, producing geometrically irregular shapes.

Third, **local population gradients**: within states, population density typically varies more smoothly than across state boundaries, where rural and urban regions may abut. State-constrained algorithms can follow these local gradients to create compact districts, while national optimization must bridge sharp density discontinuities at borders, necessitating irregular shapes to achieve population balance.

These mechanisms suggest that federalist redistricting structures—far from being geometric obstacles—provide beneficial organization for optimization algorithms. This finding has implications beyond congressional redistricting: any multi-jurisdictional partitioning problem may benefit from respecting existing boundaries that align with natural subdivisions of the underlying space.

5.2 Fractional Representation: A Theoretical Alternative

The prevalence of cross-state districts in national optimization (41.4%) raises the question of whether fractional representation could serve as a constitutional alternative to current apportionment. Under such a system, states would not receive integer seat allocations; instead, congressional districts could span state boundaries, with states receiving fractional representation based on their contribution to shared districts.

Consider the example of a district drawing 60% of its population from Vermont and 40% from New Hampshire. Under fractional representation, Vermont would receive 0.6 seats and New Hampshire 0.4 seats from this district. Aggregated across all districts, each state’s total representation would equal its Huntington-Hill apportionment, but individual districts need not respect state boundaries.

This system would require constitutional amendment, as Article I, Section 2 mandates apportionment “among the several States”—language that courts have consistently interpreted to require integer allocations. The political feasibility of such an amendment is vanishingly small, given state sovereignty concerns and the supermajority requirements for constitutional change.

Yet fractional representation offers theoretical advantages. It would eliminate the “Wyoming rule” problem, where the smallest state determines the size of all other states’ delegations. It could improve population equity by allowing districts to follow natural population contours rather than political boundaries. And it might reduce gerrymandering by expanding the solution space available to redistricting commissions.

Our results, however, demonstrate that these theoretical advantages do not translate to improved geometric compactness. Even with fractional representation, national optimization produces substantially less compact districts than state-based redistricting, suggesting that the geometric benefits of federalism outweigh the flexibility of boundary-crossing districts.

5.3 Implications for Redistricting Theory

These findings contribute to ongoing debates about constraints and objectives in redistricting algorithms. The literature typically assumes that constraints—whether legal (one person, one vote), political (protecting incumbents), or geographic (county preservation)—necessarily harm optimization objectives like compactness or minority representation. Our results challenge this assumption,

demonstrating that certain constraints may actually facilitate achieving optimization objectives by providing beneficial structure to the problem space.

This insight extends beyond redistricting to other graph partitioning applications. In community detection, for instance, respecting known organizational boundaries might improve partition quality rather than impede it. In logistics and facility location, geographic constraints aligned with natural service regions could facilitate better solutions than unconstrained optimization. The key appears to be whether constraints align with underlying structure in the data—when they do, they help rather than hurt.

The non-monotonic relationship between boundary crossing penalties and outcomes (cross-state districts peak at $\beta = 1.0$ rather than decreasing monotonically) further illustrates the complexity of constrained optimization. Small penalties do not simply interpolate between unconstrained and fully constrained solutions; instead, they create qualitatively different optimization landscapes with distinct local optima. This finding suggests caution in using penalty-based methods to achieve intermediate solutions in multi-objective optimization problems.

5.4 Limitations and Scope

Several limitations qualify our findings. First, we use a single partitioning algorithm (METIS recursive bisection). Other methods—ReCom (DeFord et al., 2019), spectral partitioning, simulated annealing—might produce different trade-offs between compactness and boundary crossing. However, METIS is widely used in computational redistricting and provides a strong baseline for comparing state-based and national configurations.

Second, we analyze only the 2020 Census. Historical data (2000, 2010) might reveal different patterns as population distribution evolves. The consistency of our findings across ablation studies suggests they reflect fundamental properties of U.S. geography and population distribution, but multi-year analysis would strengthen this conclusion.

Third, our analysis is purely computational and theoretical. We do not propose fractional representation or national redistricting as practical reforms; rather, we use these counterfactuals to understand the geometric consequences of federalist redistricting structures. The political, legal, and administrative barriers to national redistricting are insurmountable under current constitutional frameworks.

Fourth, we focus exclusively on geometric compactness, ignoring other redistricting criteria: communities of interest, county/municipal preservation, minority representation, competitiveness, and partisan fairness. National optimization might perform better on some of these criteria while performing worse on compactness. A full evaluation would require multidimensional analysis across all redistricting objectives.

Finally, we do not model the political economy of national redistricting. Who would draw national districts? How would interstate conflicts be resolved? What role would state governments play? These questions have no clear answers and represent fundamental obstacles to any national redistricting scheme beyond the geometric considerations we analyze.

5.5 Future Research Directions

Our findings open several avenues for future work. First, **algorithmic variation**: comparing METIS to ensemble methods (ReCom chains, spectral partitioning) would reveal whether the compactness-federalism relationship holds across redistricting approaches. Second, **temporal analysis**: extending the analysis to 2000 and 2010 would test whether the pattern persists across census cycles and population shifts.

Third, **partial relaxation**: rather than fully removing state boundaries, researchers could explore allowing boundary crossing only for small states (Wyoming, Vermont, Delaware) to address the Wyoming rule without compromising compactness in larger states. Fourth, **alternative geographies**: applying the methodology to state legislative districts, school districts, or international electoral boundaries would test the generality of the federalism-compactness relationship.

Finally, **multi-objective optimization**: incorporating communities of interest, minority representation, and partisan fairness alongside compactness would provide a more complete picture of the trade-offs inherent in national versus state-based redistricting. Such analysis might reveal scenarios where national optimization excels on non-geometric criteria despite underperforming on compactness.

6 Conclusion

This paper quantifies the geometric consequences of federalism in congressional redistricting through a computational experiment: we removed state boundary constraints and optimized all 435 districts nationally using graph partitioning algorithms. Comparing national optimization to state-based redistricting reveals a counterintuitive finding: **state boundaries improve rather than harm district compactness**, with national optimization producing 42.5% less compact districts (Polsby-Popper scores of 0.265 vs 0.461).

This result inverts conventional assumptions about constraints and optimization. Geographic constraints are typically understood to restrict feasible solutions and harm optimization objectives. Yet in congressional redistricting, state boundaries provide beneficial structure—multi-scale decomposition, geographic coherence, smooth population gradients—that facilitates more compact districts than unconstrained national optimization. The mechanism appears to relate to how states subdivide the national population into manageable subproblems at appropriate geographic and demographic scales.

The ablation study reinforces this conclusion: no intermediate boundary crossing penalty improves outcomes relative to state-based redistricting. Testing 21 penalty values from $\beta = 0$ to $\beta = 300$ reveals that all produce worse compactness than respecting state boundaries. The relationship plateaus at $\beta \geq 50$, where 130–150 cross-state districts persist regardless of penalty strength. Even at $\beta = 300$ (a $301\times$ penalty), 146 districts still cross boundaries. This plateau demonstrates that the geometric benefits of federalism cannot be partially captured through penalty-based methods—the choice is binary: full state-based redistricting or acceptance of substantial boundary crossing.

A secondary finding reveals that 41.4% of nationally optimized districts cross state boundaries, with mean primary state fractions of 74.4%. This pattern suggests the possibility of fractional representation—a constitutional alternative where states receive non-integer seat allocations. While politically and legally infeasible, fractional representation represents a theoretically coherent system for addressing small-state inequities and improving population equity. Our results indicate, however, that such a system would come at substantial geometric cost.

These findings contribute to three literatures. First, they extend redistricting algorithm research by demonstrating that constraints aligned with underlying data structure can improve rather than impede optimization. Second, they inform normative debates about federalism and representation by quantifying geometric trade-offs of alternative institutional arrangements. Third, they provide methodological tools—national graph partitioning with boundary crossing penalties—applicable to other multi-jurisdictional redistricting problems.

Future work should extend this analysis to multiple census years (2000, 2010), alternative al-

gorithms (ReCom, spectral methods), and multi-objective optimization incorporating communities of interest, minority representation, and partisan fairness. Such analyses would reveal whether the federalism-compactness relationship persists across contexts and whether national optimization excels on non-geometric criteria.

Ultimately, this research provides empirical evidence for a theoretical claim: the federalist structure of American representation, while rooted in political and constitutional considerations, produces tangible geometric benefits. State boundaries are not merely legal constraints to be overcome, but beneficial structures that facilitate more compact congressional districts than would be achievable through unconstrained national optimization.

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